

EYE PODS - SPECIFICATION

two required - LEFT gores 18-20, RIGHT gores 2-4 (mirror) - rows F to J - black ripstop

THE POD

Shape	shallow teardrop blister - ONE cut layer, sewn straight to the envelope at the rim
Layers	envelope gore panels + Class 3 1/2 in tape at the rim + black outer skin. There is no backing layer.
Footprint	7.96 x 6.81 m, rim 23.18 m - a MARKED OUTLINE on the envelope, not a piece
Dome height	1300 mm proud of the envelope at the apex
Position	seam to seam across three gores; row F bottom to row J top exactly
Feed holes	ENVELOPE only, each hem over one layer of Class 3 1/2 in tape: gores 20/2 panel H +450 mm; gores 19/3 panel H +910 mm; gores 18/4 panel i +850 mm

OUTER SKIN - how the dome is made

Method	the footprint outline OFFSET OUTWARD 224 mm all round (NOT a scaled copy)
Area	39.50 m2 = 115.5 % of the footprint
Why	a 1300 mm spherical cap over this footprint needs 115.5 % - the offset is solved to match
Ease	rim 24.43 m against 23.18 m = 1.24 m (5.4 %), 78 mm per segment

HOW THE PATTERN IS SUPPLIED

Method	LOFTED - offsets from a straight baseline. No giant paper template is needed.
Why	the piece is 8.0 m across; printing it 1:1 takes ~70 m2 of paper that is 95 % blank
Sheet 4	the FOOTPRINT ordinates - the outline you MARK on the envelope for the rim tape
Sheet 5	the SKIN ordinates - the whole shape, to check the assembled layer against
Sheets 6+	the skin split into bolt-width pieces; the 1:1 templates are a separate file
Stations	whole shape every 250 mm; individual pieces every 200 mm
Seam allowance	the tables give the CUT line; the SEW line is 25 mm inside - see the edge detail on sheet 3

CUTTING IT FROM 60 IN CLOTH - sheets 6 and 7

The problem	cloth is 1,524 mm wide and the table 3,658 mm long. The skin is 8.41 m across.
The answer	the skin is cut as STRIPS 1,400 mm finished, parallel to the baseline, split into sections
Outer skin	14 pieces in 4 strips - 2 are plain rectangles needing no lofting
Every piece prints	none exceeds 1450 x 3450 mm, so all 14 fit a 60 in roll 1:1
Templates	alien-balloon-eye-patterns.pdf - one piece per page, every edge labelled
Not a redesign	re-sewn at 25 mm the assembly is identical to the one-piece pattern

MATERIALS PER POD

Black ripstop	outer skin only - 39.5 m2 (the envelope is the base, so nothing else is cut)
As 60 in cloth	~37 running metres per pod, ~73 m for both - one piece per width, no nesting
Webbing	Class 3 1/2 in (12.7 mm) load tape - rim 23.2 m per pod, plus 4.7 m in the three feed-hole hems
Thread	V-69; hot knife every edge

The pod used to carry a black backing layer under the skin. It was dropped: it has no structural job - the rim tape carries the load - and it cost 34 m2 and 14 pieces per pod.

Earlier issue of this sheet scaled the skin 115 %: 132 % of the footprint area and 15 % rim ease (3.4 m to gather) - it would pucker, not dome.

[Sheet 2 build sequence](#) - [sheet 3 lofting and edge detail](#) - [sheet 4 footprint \(marking\) ordinates](#) - [sheet 5 skin ordinates](#) - [sheet 6 piecing plan](#) - [sheets 7+ piece ordinates](#)

EYE PODS - BUILD SEQUENCE

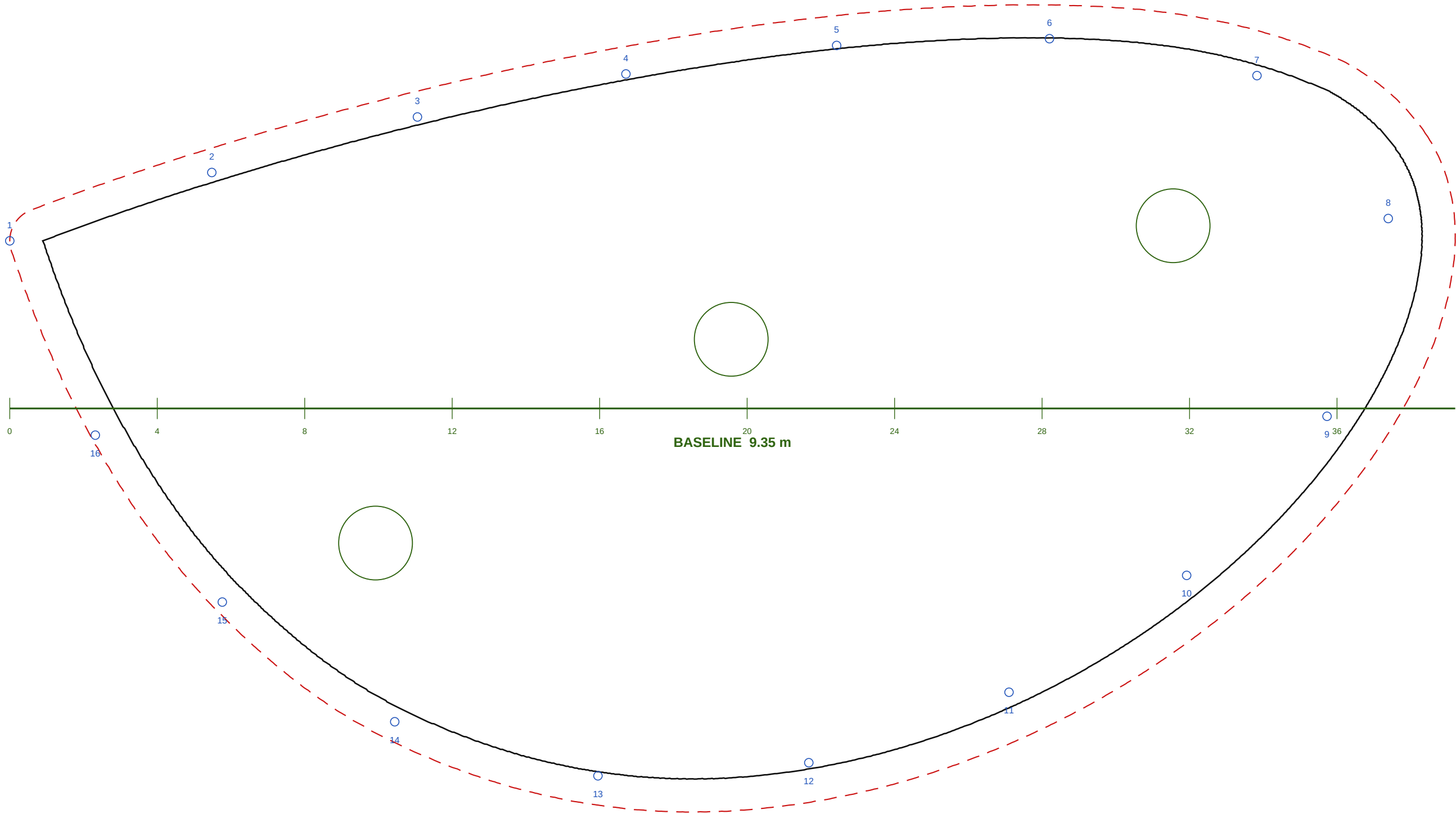
do both pods together; the RIGHT pod is the mirror of every piece

1	CUT THE SKIN	Print alien-balloon-eye-patterns.pdf at 100 % - 14 pages, one per piece, all 1:1. Hot-knife every edge. 14 pieces per pod; cut each twice, mirrored, for the two pods. Nothing else is cut: the envelope's own panels are the base of the pod.
2	SEW THE STRIPS	Join the sections within each strip first (25 mm French fell), then join strip to strip. Work bottom strip upward and keep the straight seams truly straight - they are the datum for everything after.
3	CHECK AGAINST THE LOFT	With the skin assembled, snap the full 9.35 m baseline on the floor and check the finished outline against sheet 5 every 250 mm. Correct now, not later.
4	MATCH POINTS	Mark the 16 match points from the table on sheet 3 on the assembled skin AND on the outline marked on the envelope. They are what stop the surplus skin pooling in one place.
5	TRIM AND LABEL	Confirm the assembled skin is 224 mm proud of the marked footprint all round. Label each skin LEFT or RIGHT and mark the outboard direction - the pods are mirror images, not copies.
6	FEED HOLES	Cut the three feed holes in the ENVELOPE panels, not in the pod: gores 20/2 in panel H, 450 mm up; gores 19/3 in panel H, 910 mm up; gores 18/4 in panel i, 850 mm up. O450 hot-cut, double-folded to O500 finished.
7	MARK THE ENVELOPE	With the gores joined and flat, mark the footprint outline on the envelope from the red eye outline on the F-J panel templates, and mark the same 16 points round it. This outline is the pod's base - there is no separate backing piece.
8	RIM TAPE	Sew 23.2 m of Class 3 1/2 in tape round the outline marked on the ENVELOPE, twin-needle. The tape, not the panel cloth, carries the pod loads; catch it into the vertical load tapes where it crosses a gore seam. Lap the join 150 mm.
9	PIN AND SEW	Pin the skin to the taped outline matching numbers 1 to 16: each segment holds 78 mm of surplus, worked in evenly. Sew at 25 mm, then a second row 6 mm out. Leave one segment open.
10	CLOSE AND TEST	Close the last segment, then pull 25 kg at four points. Ground-inflate: about 1300 mm proud and evenly full.

Fit the pods AFTER the gores are joined and BEFORE the parachute is rigged.

EYE PODS - LOFTING THE PATTERN

one baseline, stations every 250 mm, two offsets per station - tables on sheets 4 and 5

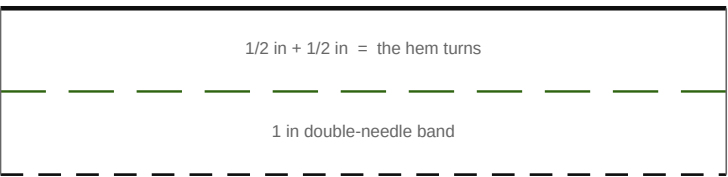


BLACK = footprint cut line RED dashed = outer skin cut line GREEN circles = feed holes (cut in the envelope)

blue numbers = the 16 rim match points - coordinates in the table below, measured in the same baseline frame

RIM EDGE DETAIL (enlarged) - where the skin meets the envelope

MATCH POINTS - baseline coordinates (mm)



CUT LINE - the ordinate tables give THIS line

FOLD LINE - 1 in inside the cut edge

FINISHED RIM - 2 in inside the cut edge

2 in (50.8 mm) outboard of the calculated appendage edge: 1/2 in turn, 1/2 in turn again, then a 1 in double-needle band.

The strip and section seams inside the skin keep the ordinary 25 mm allowance.

y is measured from the baseline: + above, - below. Feed holes: (2480, -913); (4892, +469); (7889, +1239)

no	x	y	no	x	y
1	0	+1137	9	8932	-53
2	1370	+1600	10	7980	-1132
3	2764	+1977	11	6776	-1924
4	4178	+2268	12	5418	-2402
5	5607	+2462	13	3988	-2491
6	7050	+2507	14	2610	-2125
7	8457	+2257	15	1441	-1313
8	9347	+1288	16	580	-181

EYE POD - FOOTPRINT ORDINATES (MARK ON THE ENVELOPE)

baseline 9.35 m, stations every 250 mm - NOTHING IS CUT TO THIS LINE

stn	x mm	UPPER cut	LOWER cut
0	0	+1137	+1136
1	250	+1229	+479
2	500	+1318	-40
3	750	+1404	-463
4	1000	+1486	-818
5	1250	+1562	-1116
6	1500	+1639	-1371
7	1750	+1711	-1590
8	2000	+1781	-1778
9	2250	+1848	-1940
10	2500	+1913	-2072
11	2750	+1974	-2183
12	3000	+2031	-2276
13	3250	+2086	-2355
14	3500	+2139	-2414
15	3750	+2189	-2460
16	4000	+2236	-2492
17	4250	+2281	-2508
18	4500	+2321	-2512

stn	x mm	UPPER cut	LOWER cut
19	4750	+2360	-2500
20	5000	+2394	-2473
21	5250	+2424	-2435
22	5500	+2450	-2385
23	5750	+2475	-2321
24	6000	+2491	-2244
25	6250	+2505	-2154
26	6500	+2513	-2054
27	6750	+2513	-1937
28	7000	+2508	-1808
29	7250	+2494	-1662
30	7500	+2474	-1499
31	7750	+2439	-1318
32	8000	+2391	-1113
33	8250	+2326	-884
34	8500	+2242	-622
35	8750	+2134	-317
36	9000	+1958	+58
37	9250	+1638	+595

This is where the rim tape goes on the envelope. The same outline is printed in red on the F to J panel templates.

offsets are PERPENDICULAR to the baseline: + above, - below. On rim edges the finished line is 50.8 mm inside the cut line; on strip seams it is 25 mm.

EYE POD - OUTER SKIN ORDINATES (CUT LINE)

same baseline and stations - CUT line, 50.8 mm (2 in) outboard of the calculated skin edge

stn	x mm	UPPER cut	LOWER cut
0	0	+1431	+366
1	250	+1521	-154
2	500	+1610	-579
3	750	+1694	-943
4	1000	+1774	-1250
5	1250	+1851	-1512
6	1500	+1926	-1738
7	1750	+1997	-1940
8	2000	+2066	-2110
9	2250	+2131	-2254
10	2500	+2194	-2376
11	2750	+2255	-2478
12	3000	+2312	-2565
13	3250	+2368	-2638
14	3500	+2419	-2696
15	3750	+2470	-2739
16	4000	+2515	-2770
17	4250	+2560	-2785
18	4500	+2599	-2787
19	4750	+2638	-2774

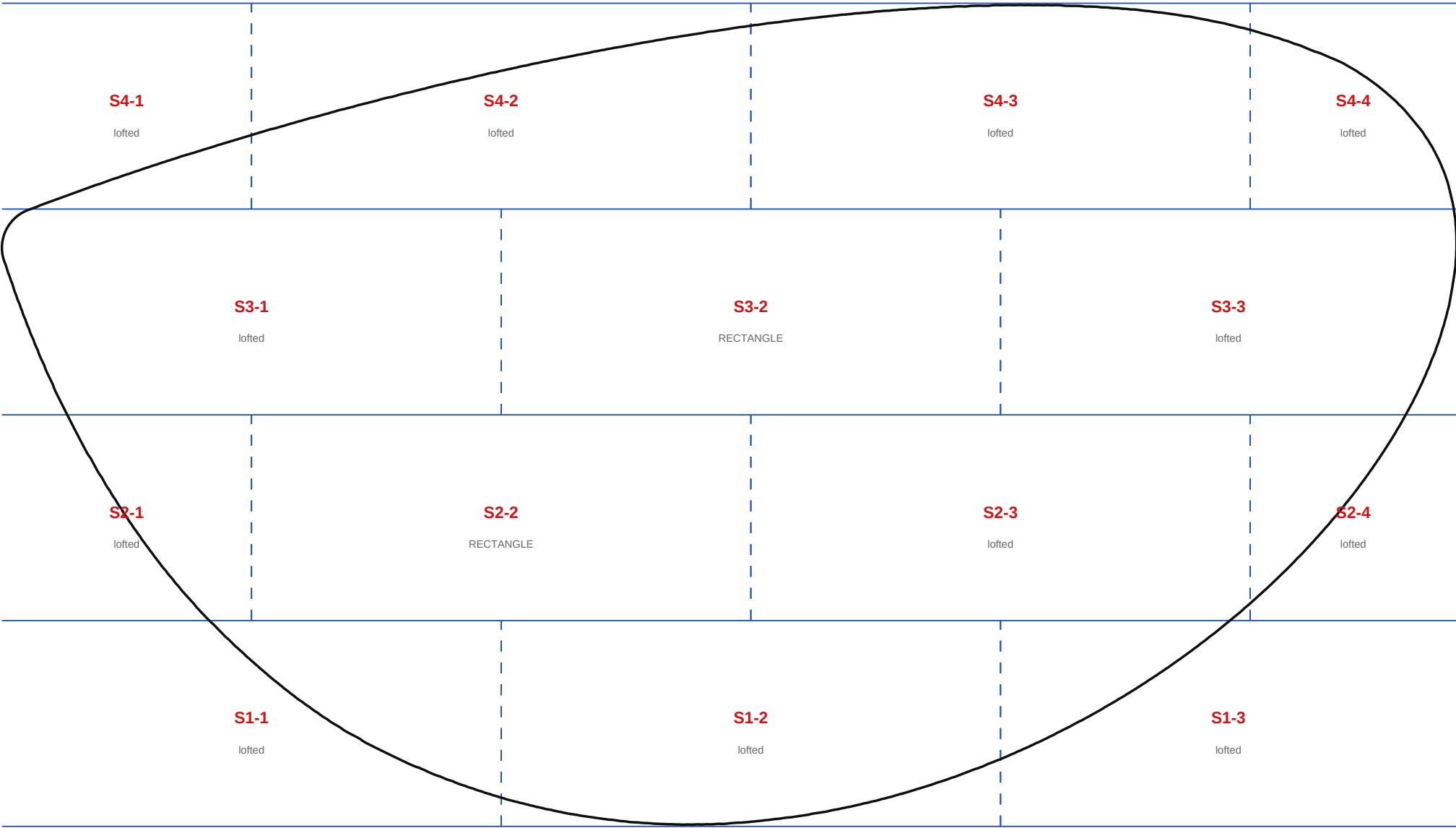
stn	x mm	UPPER cut	LOWER cut
20	5000	+2670	-2749
21	5250	+2701	-2713
22	5500	+2726	-2666
23	5750	+2749	-2606
24	6000	+2766	-2533
25	6250	+2779	-2449
26	6500	+2787	-2352
27	6750	+2787	-2242
28	7000	+2783	-2119
29	7250	+2772	-1981
30	7500	+2750	-1829
31	7750	+2718	-1659
32	8000	+2672	-1471
33	8250	+2612	-1260
34	8500	+2535	-1024
35	8750	+2439	-755
36	9000	+2304	-442
37	9250	+2100	-60
38	9500	+1752	+476

This piece is deliberately bigger: 1.24 m of rim surplus, eased in over the 16 match points.

offsets are PERPENDICULAR to the baseline: + above, - below. On rim edges the finished line is 50.8 mm inside the cut line; on strip seams it is 25 mm.

EYE POD - OUTER SKIN PIECING PLAN (x2, mirrored)

9903 x 5576 mm cut size - 14 pieces in 4 strips - the ONLY cloth cut for the pod



BLACK = pod outline (already the cut line) BLUE solid = strip seam, a straight chalk line BLUE dashed = section joint, staggered so no four seams meet

CUT LIST - sizes INCLUDE the 25 mm allowance added on internal seams

piece	cut w x h mm	pattern	add 25 mm on	piece	cut w x h mm	pattern	add 25 mm on
S1-1	2012 x 1229	ordinates	R+T	S3-1	3424 x 1450	ordinates	R+B+T
S1-2	3450 x 1413	ordinates	L+R+T	S3-2	3450 x 1450	rectangle	L+R+B+T
S1-3	1586 x 968	ordinates	L+T	S3-3	3129 x 1450	ordinates	L+B+T
S2-1	1279 x 1450	ordinates	R+B+T	S4-1	1528 x 529	ordinates	R+B
S2-2	3450 x 1450	rectangle	L+R+B+T	S4-2	3450 x 1272	ordinates	L+R+B
S2-3	3450 x 1450	ordinates	L+R+B+T	S4-3	3450 x 1413	ordinates	L+R+B
S2-4	1083 x 1333	ordinates	L+B+T	S4-4	1411 x 1244	ordinates	L+B

14 pieces, 40.8 m2, about 37 m of 60 in cloth per pod.

every piece fits the 1524 mm bolt across and the 3658 mm table along - widest 1450 mm, longest 3450 mm

L=left R=right B=bottom (the strip seam below) T=top. Edges NOT listed are pod outline - their allowance is already in the ordinates.

EYE POD - PIECE ORDINATES

each piece has its OWN baseline: the bottom edge of its finished rectangle, x = 0 at the left end. sheet 1 of 2

S1-1cut 2012 x 1229 mmadd 25 mm on R+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1204	1204	1200	1204	307
200	1204	1011	1400	1204	213
400	1204	833	1600	1204	131
600	1204	673	1800	1204	60
800	1204	535	1987	1204	0
1000	1204	414			

S1-2cut 3450 x 1413 mmadd 25 mm on L+R+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1388	183	1800	1388	30
200	1388	131	2000	1388	54
400	1388	87	2200	1388	86
600	1388	52	2400	1388	127
800	1388	26	2600	1388	175
1000	1388	10	2800	1388	231
1200	1388	1	3000	1388	295
1400	1388	3	3200	1388	368
1600	1388	13	3400	1388	445

S1-3cut 1586 x 968 mmadd 25 mm on L+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	943	0	1000	943	531
200	943	88	1200	943	666
400	943	186	1400	943	813
600	943	290	1561	943	942
800	943	406			

S2-1cut 1279 x 1450 mmadd 25 mm on R+B+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1400	1399	800	1400	182
200	1400	1029	1000	1400	0
400	1400	710	1200	1400	0
600	1400	429	1254	1400	0

S2-3cut 3450 x 1450 mmadd 25 mm on L+R+B+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1400	0	1800	1400	0
200	1400	0	2000	1400	0
400	1400	0	2200	1400	0
600	1400	0	2400	1400	0
800	1400	0	2600	1400	0
1000	1400	0	2800	1400	0
1200	1400	0	3000	1400	0
1400	1400	0	3200	1400	0
1600	1400	0	3400	1400	117

S2-4cut 1083 x 1333 mmadd 25 mm on L+B+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1283	0	800	1283	875
200	1283	186	1000	1283	1181
400	1283	389	1058	1283	1282
600	1283	616			

UPPER and LOWER are measured UP from the piece's baseline - both are positive. Where a value is constant that edge is a straight seam; where it changes it is the pod outline.

Straight seam edges: mark the sew line from the table, then add 25 mm outside it. Curved pod-outline edges: the table IS the cut line.

EYE POD - PIECE ORDINATES

each piece has its OWN baseline: the bottom edge of its finished rectangle, x = 0 at the left end. sheet 2 of 2

S3-1cut 3424 x 1450 mmadd 25 mm on R+B+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1155	1119	1800	1400	0
200	1400	542	2000	1400	0
400	1400	89	2200	1400	0
600	1400	0	2400	1400	0
800	1400	0	2600	1400	0
1000	1400	0	2800	1400	0
1200	1400	0	3000	1400	0
1400	1400	0	3200	1400	0
1600	1400	0	3399	1400	0

S3-3cut 3129 x 1450 mmadd 25 mm on L+B+T

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1400	0	1800	1400	0
200	1400	0	2000	1400	0
400	1400	0	2200	1400	0
600	1400	0	2400	1400	0
800	1400	0	2600	1400	0
1000	1400	0	2800	1400	76
1200	1400	0	3000	1400	551
1400	1400	0	3104	1210	1121
1600	1400	0			

S4-1cut 1528 x 529 mmadd 25 mm on R+B

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	0	0	1000	349	0
200	75	0	1200	411	0
400	148	0	1400	472	0
600	216	0	1503	503	0
800	284	0			

S4-2cut 3450 x 1272 mmadd 25 mm on L+R+B

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	504	0	1800	961	0
200	561	0	2000	1004	0
400	619	0	2200	1044	0
600	674	0	2400	1083	0
800	725	0	2600	1119	0
1000	776	0	2800	1155	0
1200	825	0	3000	1187	0
1400	873	0	3200	1218	0
1600	919	0	3400	1247	0

S4-3cut 3450 x 1413 mmadd 25 mm on L+R+B

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1248	0	1800	1387	0
200	1273	0	2000	1387	0
400	1298	0	2200	1382	0
600	1320	0	2400	1372	0
800	1338	0	2600	1358	0
1000	1354	0	2800	1336	0
1200	1367	0	3000	1307	0
1400	1379	0	3200	1268	0
1600	1383	0	3400	1220	0

S4-4cut 1411 x 1244 mmadd 25 mm on L+B

x mm	UPPER	LOWER	x mm	UPPER	LOWER
0	1219	0	800	889	0
200	1160	0	1000	725	0
400	1087	0	1200	484	0
600	1007	0	1386	4	0

UPPER and LOWER are measured UP from the piece's baseline - both are positive. Where a value is constant that edge is a straight seam; where it changes it is the pod outline.

Straight seam edges: mark the sew line from the table, then add 25 mm outside it. Curved pod-outline edges: the table IS the cut line.